ANT User Evaluation Script

Goal: See whether people with disabilities can use ANT on their own, and where it fails them. We test the app, not the person. **Setting:** Seated at a table in an office. Nobody walks. **Time:** 55 min. **Team:** facilitator and note-taker (plus an interpreter or caretaker if the participant wants one).

Before the Session

- Phone charged, tester build installed, internet and location on.
- A team member's phone as the **test caretaker**. Never use a real caretaker.
- Props on the table: 3 bus photos (different numbers), 2 signs (Bangla, English), 1 obstacle photo.
- Mock-GPS app running a test route (Task 3). Street-noise recording on a laptop (Tasks 2, 4, 6).
- Consent forms, score sheets, recording on. Use IDs (P01, P02), never names.

Start (read aloud)

- "We are students from United International University working with BUET. We built an app to help people travel safely. Today you try it and tell us what works and what doesn't."
- "**You cannot fail.** If something is confusing, the app is wrong, not you."
- "We would like to record audio, video and the phone screen. Only our team sees it. Your name stays private. Is that okay?"
- "You can stop, take a break or skip any task at any time."
- "Some tasks pretend to be an emergency. **No real alarm will be sent.**"

Note-taker records: ID, age range, disability type, mobility aid, language (English/Bangla), smartphone use, screen reader use.

Demo and First Impressions (about 10 min, before any task)

Demo (facilitator, 3 min). Show the app working, using examples that are **not** in the tasks below so we don't prime the participant: ask it for a route to a landmark, point the camera at an object and hear the description, and show the "Show my screen" message board. Describe aloud what you do. Hand over the phone and let them touch it freely for a minute.

High-level feedback (ask each, then "why?"):

- "What do you think of it?"
- "What do you **like** about it? Why?"
- "What do you **dislike** about it? Why?"
- "What would you **change**? Why?"

Record the answers word for word. Don't defend or explain. Then say: "Now we will try some situations. Remember, you cannot fail."

How Each Task Works

1. Facilitator reads the scenario. **Never say which button to press.**
2. Participant tries. Ask them to say what they think (don't force it).
3. If stuck for 30 s, help step by step: **L1** "What would you try next?" → **L2** point to an area → **L3** tell the exact action.

4. Score it, then ask: **"How easy was that?"** 1 (very hard) to 5 (very easy). Use a face card; blind participants answer aloud.

Score	Meaning
Pass	Done alone
Assisted	Done with L1-L2 help
Fail	Needed L3, gave up, or the app gave a wrong or unsafe result

Always write down: **time**, **errors**, and the **exact words** said to the app.

Tasks

#	Say to the participant	Done when
1	"Set up the app the way you need it."	Language, vision, text size and mobility aid are right
2	"Ask the app to take you to ____." (their own place or a saved one) Then: "Now ask for **the market**."	A route to the right place starts. For "the market" the app asks a question, it does not guess
3	"Pretend you are walking. Listen." *(Team moves the mock GPS.)* Ask: "Where are you? What is the next turn?"	They answer both correctly
4	"Change your mind. Stop the trip." Then: "Go somewhere else."	The trip stops. A new route starts
5	*(Hold the prop up, 1-2 m away.)* **A** "Which bus is this?" **B** "What does this sign say?" **C** "Is anything in your way?"	The answer matches the prop. **A wrong answer is a Fail**, even if they don't notice
6	"Pretend someone is following you and you are scared. Tell the app." Then: "Oops, that was a mistake. Stop it."	The alarm starts without the word "help". It cancels within 5 s. The test caretaker phone gets the right message
7	"Connect your caretaker." *(Give the test code.)* Then: "Tell them you are okay."	They connect alone. The caretaker gets "okay", not a generic alert
8	"Without speaking, ask me which bus this is." *(Show my screen)*	You can read it from arm's length
9	"You see a pothole. Warn other people."	They reach the Submit screen with the right type. **Do not press Submit**

Mute participants: type or use buttons instead of speaking in Tasks 2, 4 and 6. Note every task that needed speech.

Final Questions

Rate 1 (No) to 5 (Yes): I could use this alone. · It understood me. · Its answers were clear. · I felt safe. · I would use it on a real trip.

Then ask:

- Has your opinion changed since the demo? How?
- Best thing? Worst thing?

- When did you feel lost, scared or confused?
- Did the app ever say something wrong?
- What is missing for a real trip?
- Were the voice, text size and language right?

How Helpful Is It? (each feature)

Go through the features one at a time: **route guidance, camera, emergency button, caretaker, message board, hazard report**. For each ask:

- "How **helpful** would this be in your daily life? Why?"
- "How **useful** is it compared with what you do today? Why?"
- "When would you use it? When would you not?"

Then: "Which feature would help you **most**? Which **least**? Why?"

What Would You Change? (open questions)

Ask each. Then ask "why?" and write their words down.

- "What **changes** would you want to make to the app?" (Let them list as many as they like. Ask "anything else?" until they stop.)
- "Pick one feature: route, camera, emergency, caretaker, message board, hazard report. How would **you** make it work?"
- "What should the app do that it does not do now?"
- "What would stop you using it every day?"
- "What worried you, or felt risky? What would make you trust it more?"
- "How should it talk to you? Speed, voice, length of answers?"
- "What would you remove?"
- "What would you tell a friend with the same needs?"
- "Describe the app in three words."

Thank them. Confirm they are happy for the recordings to be used.

Adjust for the Participant

Group	Do this
Blind	All oral. Describe the room and props first. Say before anything moves or makes sound.
Low vision	Check text size and contrast first. Large print cards.
Deaf	Written cards or interpreter. Note whether visual or vibration alerts are enough.
Mute	Written or typed answers.
Cognitive	Short sentences, one step at a time, read each task twice, face cards, extra time. Stop at the first sign of stress.
Wheelchair	Check table height, knee room and one-handed use.

Rules: Don't explain or defend the app. Wait 10 s before helping. Don't lead ("What did you expect?", not "Wasn't that easy?"). If the participant is distressed, stop. Safety before data.

Score Sheet (one per participant)

Task	Pass / Assisted / Fail	Time (s)	Errors	Ease (1-5)	Exact words / notes
1 Setup					
2 Destination					
3 Route					
4 Change mind					
5A Bus					
5B Sign					
5C Ahead					
6 Emergency					
7 Caretaker					
8 Show screen					
9 Hazard					

Also record the open answers for each participant.

After all sessions: per task, report Pass / Assisted / Fail counts, average ease and time, split by disability group. A Fail on Task 5, 6 or 7 is a **safety issue** and goes first. Rank other problems by how many people hit them × how serious they were.

State in every report: this was a seated office test. Real walking, traffic, outdoor GPS, glare and moving buses were not tested.